

Project Milestone 2

Milestone	Brief Description	Deadline
2: Formulation (5%)	Two-page literature survey of minimum 10 relevant research articles in the 5 preceding years (2017 – 2021) and Problem Formulation of the assigned project.	Week 5: Tuesday, 2 November 2021

Milestone 2 Objectives and Requirements:

1. Literature Review:

- Write a comprehensive literature review of 10 relevant and recent scientific articles with a publication date between 2017 and 2021.
- These articles should be relevant to your specific project topic and should implement different meta-heuristic optimization techniques on your topic.
- It is preferable that the articles are under respected publishers (e.g. IEEE, Elsevier, etc.).
- The review should discuss the recent advances in your topic as well as highlight the accomplishments and shortcomings of each article's approach.
- The review should highlight important results in literature for problem instances or benchmarks, to which the project future results can be compared and validated.
- The review should identify any metrics upon which the performance of the algorithm can be measured on your respective problem.

2. Problem Formulation:

- Formulate your specific multi-objective problem clearly – in-text and mathematically.
- Identify the decision variables, the objective functions (**min. 2 objectives**) and the constraints (**min. 3 constraints**) – providing the mathematical expressions of the objective and constraints functions.
- Discuss how this mathematical formulation relates to your project statement and objectives.
- The problem formulation should be general for any problem size.
- State clearly any assumptions made for your problem.

Submission Deadline:

- The deadline of Milestone 2 submission is on **Tuesday, 2 November 2021, at 23:59**.

Deliverables:

- Progress report (in provided template below) containing the literature review (max. 2 pages) and the problem formulation in accordance with the objectives and requirements above.
- Report Template link: <https://www.overleaf.com/read/qvbjycgzhmmb>

The milestone will be graded based on several criteria including **but not limited to**:

- Quality of all submitted material.
- For reporting: clarity of presentation and professional discussion with multiple aids (e.g. block diagrams, flowcharts, figures, tables, graphs, charts, etc.). Equations should be written professionally with proper identification of nomenclature.